OFFLINE PIPELINE

Offline Agentic IFC-To-Navigation-Priors Complier [This Work]

1 IFC File



Input building model as IFC file

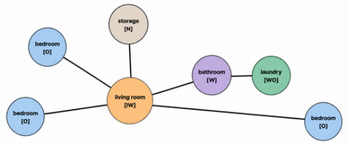
2 LLM Extraction

LLM Agent

Extracts spatial, semanantic and robot-relevant information using an LLM (Gemini Flash 3.0)

3 Enriched Semantic Graph





Builds a multi-layered semantic graph with priors for navigation and affordance planning

4 Multi-Layer Navigation Graph

- SEMANTIC LAYER
- TOPOLOGICAL LAYER
- AFFORDANCE LAYER
- METRIC LAYER
- . . .

Defines the multi-layer navigation contract between offline complier and online agent

Multi-Layer Navigation Contract

ONLINE PIPELINE

Online Robot Navigation & Replanning [Future Work]

1 MCP-ROS Bridge

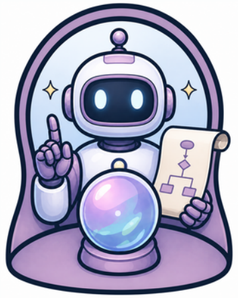
ROS2

Bridges BIM retrieval to ROS 2 compatible formats via MCP

Waypoints

Status / Failure Reports

2 Agentic Navigation Planner



Trajectory Planning
Alternative Re-Routing
Evaluate Affordances
Update Model

Agentic Robot LLM Planner and Navigation Assistant that reasons over BIM to generate waypoints and plans

Nav2 Goals

Feedback

3 Robot Agent

Nav2 Waypoints

Reasoning & Feedback



Executes plan, provides feedback from sensor data and gives live feedback